DEL-Bench: Evaluating LLM Higher-Order Belief Tracking with Dynamic Epistemic Logic

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Abstract

Multiagent LLM systems will eat the world. At least that's the promise. More and more, workflows require an agent to reason about other agents — what they believe, and sometimes what they are deliberately hiding. Most tests of this ability still lean on hand-written answer keys or informal judgments about what an agent ought to believe. To combat this, we introduce DEL-Bench[1], a benchmark for evaluating the higher-order belief tracking of LLMs against truths generated by a dynamic epistemic logic (DEL) kernel based on the work by Baltag and Smets. The benchmark uses a family of coin-game scenarios that are organized along four difficulty axes, which aim to test and lay bare different shortcomings in the LLMs reasoning. Each scenario is formulated both as a formal epistemic plausibility model which can be solved by the kernel, and as a natural-language. Each prompt is repeated and paraphrased twice for consistency and statistical rigour.

We test five LLMs from different providers and sizes, using both direct and chain-of-thought prompting, across 17 scenarios and 140 probes. We also include a small separate check for conditional belief revision. The main challenge is not sentence comprehension: models score nearly perfectly (99.4%) on syntactic depth-3 controls (controls where the sentence structure is complicated but it is logical depth is not) but their accuracy drops to (69.6%), with further degradation when belief nesting is added. Because gold answers come from formal models, DEL-Bench reveals failures that simple accuracy scores miss. For matched scenario pairs that differ only epistemic structure, models give the same answer in both cases in up to 28% of cases indicating that LLMs often focus on facts in the prompt instead of tracking who knows or believes what. These findings suggest that DEL can provide a strong formal basis for diagnostic LLM benchmarks focused on nested belief, knowledge, and deception.

1 Introduction

Multi-agent LLM systems are now used to automate many tasks, of which a growing percentage requires collaboration between the individual agents [Guo et al., 2024]. These settings require more than just answering questions that have clear cut answers. Agents need to who knows what, be able to spot deception, judge the trustworthiness of a source, and track even higher-order beliefs. Mistakes here cannot be ignored as nuisance as they can cause information gaps and disrupt group dynamics in decentralized LLM systems, leading to critical failures as agents become more independent [Hammond et al., 2025].

Dynamic epistemic logic (DEL) is a prominent framework for modelling changes in knowledge and belief in dynamic, multi agent scenarios, making it a suitable framework for constructing LLM belief-tracking evaluations. The new approach presented in this project is to use DEL as a ground-truth generator, rather than modelling the LLM as a DEL agent with an internal plausibility ordering. Each benchmark item comprises a formal model, a natural-language rendering, a probe question, and a set of answer choices.

[1] `https://github.com/LeoMetasch/DEL-Project`

In this paper, we introduce DEL-Bench, a benchmark designed to test higher-order belief tracking in LLMs. The benchmark includes 17 scenarios and 140 probes, all based on the coin example from Baltag and Smets's qualitative theory of dynamic interactive belief revision [Baltag and Smets, 2008]. Scenarios are grouped along four difficulty axes: observation, deception, trust, and asymmetry. DEL-Bench is meant to go beyond simple tests of whether LLMs can answer basic questions, but also if they can track nested beliefs and knowledge in both public and private situations, including cases involving deception.

Our main contributions are:

1. We implement a DEL kernel for epistemic plausibility models, formula evaluation, and the Action-Priority product update.

2. We construct a benchmark of natural-language scenarios with ground-truth answers are generated from formal DEL models.

3. We introduce scenario pairs, comprehension controls, and metrics that distinguish language comprehension from genuine belief tracking.

4. We establish a baseline result, testing five LLMs, using two prompting strategies, across three paraphrases, showing that logical depth remains the major source of error.

2 Previous Work

LLM-based agents are no longer only a speculative interface pattern. Recent surveys describe a rapidly growing ecosystem consisting of autonomous and multi-agent LLM systems autonomous problem solving and guidance on decision making [Guo et al., 2024, Li et al., 2024]. Frameworks such as AutoGen make the design of systems in which multiple LLMs agents communicate, trivial, and lower the entry barrier to letting LLMs make decisions with limited human involvement [Wu et al., 2023]. Generative-agent simulations show how LLMs can be embedded in persistent agents with memory, and the ability to plan and delegate [Park et al., 2023]. More recent surveys highlight that communication is the central design object in such systems. Agents must make decisions regarding trust, secrecy and collaboration/deferment, and how to coordinate given incomplete or conflicting information [Yan et al., 2025]. This trend also points to agent systems offered as services, where multiple semi-autonomous components interact with other services and users in ways that are not directly supervised at every step [Li et al., 2024].

This interaction creates risks. Multi-agent AI systems often fail through miscoordination or conflict. They collude with other agents, or fall victims to information asymmetries and network effects. These destabilizing dynamics can lead to multi-agent security failures [Hammond et al., 2025]. On the individual behaviour level, LLMs are known to exhibit sycophancy, conforming with a user's stated views and opinions rather than staying objectively accurate [Sharma et al., 2025]. Inside multi-agent settings, the corresponding behaviour is conformity. Models shift toward peer messages or group positions even when the corresponding information should not be treated as decisive [Weng et al., 2025, Choi et al., 2025]. Population-level experiments show that LLM agents can develop shared conventions and collective biases, and that committed adversarial minorities can push a larger LLM population toward alternative conventions [Flint Ashery et al., 2025]. Security work mirrors the same concern from an adversarial perspective. Malicious or compromised agents can hide their intentions [Xie et al., 2025], and adversarial content can hijack communication in multi-agent systems, especially those with limited human supervision, in some cases leading to code execution or data exfiltration [Triedman et al., 2025].

These risks make higher-order belief tracking more than a theoretically curious cognitive benchmark. When dealing with adversarial interaction, an agent must distinguish what is true from what another agent says, what that speaker knows, what the speaker wants listeners to believe, and what other agents believe about the speaker's beliefs. Deception is precisely a mismatch between an what was said, the world, and the speaker's own information state. A system that cannot represent this structure is vulnerable to treating persuasive arguments as fact. This is why benchmarks for theory of mind and belief tracking are increasingly relevant for LLM evaluation. Several recent benchmarks attack pieces of this. FANToM stress-tests models in conversations where information is unevenly shared [Kim et al., 2023]. HI-ToM pushes on recursive higher-order belief and finds accuracy dropping as the order climbs [He et al., 2023]. SimpleToM makes a different cut, separating whether a model can infer a mental state from whether it can act on that inference to predict behaviour [Gu et al., 2024]. One additional note is that human comparison studies also show that LLM theory-of-mind performance continues to be sensitive to test design and uncertainty [Strachan et al., 2024].

Dynamic epistemic logic is a natural formal tool for this space because it was designed to model how knowledge and belief change through public observation, private observation, communication, and belief revision. Baltag and Smets's doxastic DEL framework gives a qualitative semantics for plausibility, conditional belief, safe belief, and action-based update [Baltag and Smets, 2006, 2008]. Outside the LLM setting, DEL has also been used to formalize theory-of-mind reasoning as the dynamic update of beliefs about beliefs [van de Pol et al., 2018]. This makes DEL particularly well suited to adversarial and asymmetric-information settings. It can specify not only who knows a fact, but also who believes that someone else knows, believes, or falsely believes it.

The direct intersection of DEL and LLMs is recent but is already informative. MindGames uses dynamic epistemic modal logic to generate controlled theory-of-mind problems and verbalize them as natural-language inference tasks [Sileo and Lernould, 2023]. DEL-ToM decomposes ToM tasks into DEL-grounded belief-update traces and uses automatically generated DEL data to train a verifier that scores candidate reasoning processes at inference time [Wu et al., 2025]. Obiso et al. [2025] integrate DEL with geometric vector representations to model epistemic friction in dialog. These projects show that DEL can help with two complementary goals: building formally controlled evaluation items and making belief-update reasoning more transparent. Further examples of LLM group behaviour use Bayesian social learning over dynamic logic [Jain and Krishnamurthy, 2025]. While these approaches are valuable, they do not directly provide the symbolic modal structure needed to validate nested knowledge and belief formulas.
The presented literature also clarifies the challenges that come with such evaluation frameworks. First, LLMs have parametric knowledge from their training data, which is retained in model weights, and can conflict with prompt context. This can make benchmark answers depend on memorized or specific associations [Xu et al., 2024]. Secondly, benchmark contamination is a real issue for classic theory-of-mind tasks, which lead to a shift towards synthetic and formally generated scenarios rather than relying reused human psychology items [Sileo and Lernould, 2023, Strachan et al., 2024]. Third, an LLM's answer is not always reliable evidence of an internal belief. Belief measurement in LLMs requires additional criteria including accuracy, coherence, uniformity [Herrmann and Levinstein, 2025]. DEL-Bench addresses some of these issues by making the formal DEL kernel the main scientific contribution. The kernel constructs epistemic plausibility models, applies Action-Priority updates, and computes gold answers by deterministic computation. The benchmark layer is then put on top of this kernel. Natural language prompts are generated on the basis of the underlying coin scenario, answer choices are tied to formal formulas, and model outputs are evaluated against ground truths rather than informally verified.

3 Formal Foundations

The logical foundation of DEL-Bench is that of the epistemic and action plausibility models. These models come from the line of qualitative belief-revision work developed in Baltag and Smets [2006] and Baltag and Smets [2008] and we encourage the reader to consult these papers for a more detailed explanation. The core idea is to represent doxastic attitudes not with probabilities, but by an ordering over epistemically possible worlds. This framework sufficiently serves our needs as the questions we pose are qualitative in nature: whether Alice believes that Bob knows the coin's side, whether Bob believes that Alice has been deceived, or whether an agent has no settled belief either way.

3.1 Epistemic Plausibility Models

Consider an agent that believes various different scenarios to be possible and, though unsure, has ordered them from most to least plausible. Formally, we say the agent has a plausibility model, where W is the set of possible worlds, $\sim_a$ captures which worlds agent a cannot distinguish, $\leq_a$ is a plausibility ordering over these worlds (where $w_1 \leq w_2$ means $w1$ is at least as plausible as w_2), and V assigns truth values to atomic propositions at each world. Worlds indistinguishable to the agent are grouped in *information cells* sometimes also referred to as *equivalence classes.* The *actual* world is denoted by w^* but agents do not have definite knowledge of it.

Definition 1 (Epistemic plausibility model)**.** *An epistemic plausibility model is a tuple*

$$M = (W, \{\sim_a\}_{a \in A}, \{\leq_a\}_{a \in A}, V, w^*),$$

where W is a finite set of possible worlds, A is a finite set of agents, $\sim_a$ is agent a's indistinguishability relation, $\leq_a$ is agent a's plausibility preorder, V is a valuation function for atomic propositions, and w^* is the actual world.

Knowledge and belief are defined as follows: $K_a\varphi$ (read: agent a knows φ) holds *iff* φ is true for all worlds in agents a information cell; $B_a\varphi$ (read: agent a believes φ) holds when φ is true for at least the most plausible worlds in that cell. We note that these definitions allow for the concept of mistaken belief: an agent may believe φ because all of the best worlds compatible with her information make φ true, even though φ is false at the actual world.

Beyond simple belief, the plausibility ordering lets us express what an agent would believe under a supposition. Conditional belief $B_a^{\psi}\varphi$ (read: agent a believes φ given ψ) holds when φ is true at the most plausible worlds of agent a's information cell that also satisfy ψ. Intuitively, it captures what a would come to believe were she to learn ψ: she believes the most plausible world where ψ is true.

Baltag and Smets also introduce richer doxastic notions such as safe belief - a more entrenched form of defeasible belief, namely one that persists under revision with any true information. These are beyond the scope of this paper, however, and as they are not evaluated by the benchmark, we will not go into further detail here.

3.2 Action Plausibility Models

The EPM captures what the agents believe at any given moment but it says nothing about how those beliefs change when an event takes place and the agent receives new information. To represent an event, Baltag and Smets [2008] introduce the action plausibility model. Just as agents are uncertain about what the actual world is and order the possible worlds by plausibility, they may be uncertain about which event is actually occurring and order those by plausibility too. An action model is thus defined analogously to the EPM, with worlds replaced by events and the valuations replaced by a preconditions:

Definition 2 (Action plausibility model). *An action plausibility model is a tuple*

$$\Sigma = \left(E, \{\sim_a^\Sigma\}_{a\in A}, \{\leq_a^\Sigma\}_{a\in A}, pre, e^*\right),$$

where E is a finite set of possible events, $\sim_a^\Sigma$ is agent a's indistinguishability relation on events, $\leq_a^\Sigma$ is agent a's plausibility preorder on events, pre is a precondition map assigning a formula to each event, and e^* is the actual event.

The precondition $pre(e)$ specifies the worlds in which event e can occur: e is applicable at world w exactly when $M, w \models pre(e)$.The plausibility order on events encodes the agents' beliefs about which event is taking place; for example, a soft announcement that an agent trusts is modeled by making the truthful event strictly more plausible to her than the alternatives, while a fully reliable hard announcement leaves no uncertainty about the event at all.

3.3 Action-Priority Product Update

Given an initial world model and an action model, how might the agents update their beliefs upon receiving the new piece of evidence? The merging of the two is the purview of the Action-Priority Update. It is called the Action-Priority update because the information gained from the new action is prioritized, and the initial ordering is used only as a tie-breaker.

We combine the set of worlds and events by taking the Cartesian product $W \times E$, restricted to pairs (w, e) where the world w satisfies the precondition of event e; that is, $M, w \models pre(e)$. Next, we define the new plausibility relations on these pairs via an anti-lexicographic merge. Since actions have priority, if event e_1 is strictly more plausible than event e_2, then any compatible pair (w_1, e_1) is more plausible than the corresponding pairs containing e_2. In the case where e_1 and e_2 are equally plausible, the update breaks the tie by consulting the original state model and orders the pairs according to the old world plausibility. Formally, for each agent a:

$$(w, e) \leq'_a (w', e') \iff \left(e <_a^\Sigma e' \wedge w \sim_a w'\right) \vee \left(e \simeq_a^\Sigma e' \wedge w \leq_a w'\right).$$

Baltag and Smets use this framework to simulate several familiar belief-change operations. Public announcements of hard facts delete worlds incompatible with the announcement and establish knowledge. Lexicographic upgrades represent soft but trusted evidence: worlds satisfying the new evidence become more plausible than worlds that do not, while the previous ordering is preserved inside each zone. Conservative upgrades represent weaker evidence: only the best worlds satisfying the evidence are moved to the top, while the rest of the ordering is left as intact as possible.

In the original DEL theory, these operators describe how formal agents revise their doxastic states. In DEL-Bench, the role is different but closely related. We use the Action-Priority update to construct the benchmark's formal scenario states: public observations, private observations, truthful announcements, successful lies, unreliable announcements, and soft announcements. The LLM is then asked natural-language questions about those states and asked to track beliefs and higher-order (nested) beliefs when events occur, and its answer is compared with the DEL-generated gold answer, computed deterministically by the kernel defined in Section 4. Thus, Action-Priority update is not a hypothesis about the LLM's own internal update rule. Rather it is the logical engine that makes the benchmark answers precise and consistent across all scenarios.

4 The DEL Kernel

The kernel is the core of DEL-Bench: a self-contained implementation of a part of the of the Baltag–Smets framework [Baltag and Smets, 2008], used as a ground-truth generator rather than as a model of the LLM theory of mind (ToM). It is composed of four parts — a formula language, the two model types (epistemic plausibility models and action models), the Action-Priority update computer, and a (recursive) model checker — together with a library of canonical action models and a layer of tests. The benchmark layer of Sections 5–7 is built entirely on top of these components: every scenario is assembled from kernel objects, and every gold answer is obtained by model checking a formula in a kernel-constructed model rather than by hand-authoring. The Kernel has been designed to be easily extensible and scalable. The current implementation serves merely as a foundation and we encourage adaption as needed. We describe each component in turn.

4.1 Formula Language

Each query about the coin game is a formula represented as an abstract syntax tree. Every node is one of the following six primitive types:

1. `Atom(name)`: an atomic proposition, such as the proposition that the coin shows heads.
2. `Not(`φ`)`: the negation of φ.
3. `And(`φ`, `ψ`)`: the conjunction $\varphi \wedge \psi$.
4. `K(`a`, `φ`)`: agent a knows φ.
5. `B(`a`, `φ`)`: agent a believes φ.
6. `ConditionalB(`a`, `ψ`, `φ`)`: conditional belief; given ψ, agent a believes φ.

The additional constructors `Or` and `Implies`, the tautology constant `TOP`, and the `nested` function are convenience helpers: they construct formulas using the six primitive node types rather than being nodes themselves. Ordinary belief is the special case $B_a\varphi \equiv B_a^{\top}\varphi$. These primitive constructors cab be nested without any depth limit. Safe belief, as defined by [Baltag and Smets, 2008], is not supported. Section 4.5 explains how the supported operators are evaluated.

4.2 Epistemic Plausibility Models

A possible world is represented by `World`. It stores a label and a valuation: the set of atomic propositions that are true at that world. An epistemic plausibility model is represented by `EPM`, corresponding to the tuple

$$M = (W, \{\sim_a\}_{a\in A}, \{\leq_a\}_{a\in A}, V, w^*)$$

from Section 3.1. It stores the possible worlds, the agents, and the actual world w^*.

For each agent a, the model stores two relations. The indistinguishability relation $\sim_a$ is represented as a partition of the possible worlds into information cells. Worlds in the same cell cannot be distinguished by the agent. The plausibility preorder $\leq_a$ is represented as a set of ordered pairs (s, t), where

$$s \leq_a t$$

means that s is at least as plausible as t. Thus, smaller elements are more plausible, and the most plausible worlds are the $\leq_a$-minimal ones.

The model checker has three methods available to interact with an `EPM`:

- `cell(`a`, `w`)` returns the information cell $[w]_a$ containing w;
- `leq(`a`, `s`, `t`)` tests whether $s \leq_a t$; and
- `min_plaus(`a`, `S`)` returns the most plausible worlds in a set S i.e. its $\leq_a$-minimal elements.

Within each information cell, $\leq_a$ is a well-defined preordering. Consequently, every non-empty subset of an information cell has at least one most plausible world. For the empty set, `min_plaus` returns the empty set. "'latex

4.3 Action Models

An action model describes how an event changes the agents' information and beliefs. The implementation is analagous to the `EPM` and represents individual events using `Action`. Each event has a label and a *precondition*: a formula specifying the worlds in which that event can occur.

An `ActionModel` corresponds to the tuple

$$\Sigma = \left(E, \{\sim_a^\Sigma\}_{a\in A}, \{\leq_a^\Sigma\}_{a\in A}, pre, e^*\right).$$

from Section 3.3. It stores the possible events, the agents, and the actual event e^*. For each agent, it also stores an indistinguishability partition over events and a plausibility preorder over events. Like for the worlds, a smaller element is more plausible.

There are three helper functions that enable the construction of these relations:

- `total_preorder_pairs` converts an integer ranking into a plausibility preorder. Lower ranks are more plausible, and equal ranks represent equiplausibility.
- `discrete_partition` places each event in a separate cell, representing an agent who can distinguish every event.
- `total_partition` places all events in one cell, representing an agent who cannot distinguish between any of them.

The kernel also provides constructors for the complete suite of action models used in the benchmarking scenarios:

- `public_announcement(`P`)` represents a hard public announcement $!P$. Its single event has precondition P, so worlds that do not satisfy P are removed by the update.
- `fair_game_observation(observer, `P`)` represents an observation in which the observer learns whether P is true. Other agents know that an observation occurred but cannot distinguish its outcome.
- `fully_private_observation(observer, `P`)` represents an observation hidden from the other agents. The observer learns whether P is true, while the other agents consider it most plausible that nothing happened.
- `successful_lie(speaker, `P`)` represents a public claim that may be a honest or a lie. The listeners cannot distinguish the two cases but consider honesty more plausible. By default, the lie event is the actual one.
- `lexicographic_upgrade(`P`)` represents a soft, trusted announcement $\Uparrow P$. It promotes P-worlds above $\neg P$-worlds without removing either group.
- `unreliable_announcement(`P`)` represents a completely untrusted announcement. The possible outcomes are equiplausible, so the prior belief ordering is preserved.

4.4 Action-Priority Product Update

The function `product` combines an `EPM` with an `ActionModel`. It returns the updated epistemic plausibility model defined by the Action-Priority product update of Section 3.3.

The first step constructs the updated worlds:

$$W' = \{(w, e) \in W \times E \mid M, w \models pre(e)\}.$$

Thus, a world–event pair survives exactly when the event's precondition holds at that world. Each surviving pair inherits its valuation from w, since the implemented events change beliefs rather than atomic facts. The actual world of the updated model is (w^*, e^*). If this pair is not executable, the function raises `ActionInapplicableError`.

The new indistinguishability relation is

$$(w, e) \sim'_a (w', e') \quad \Longleftrightarrow \quad w \sim_a w' \ \wedge \ e \sim^{\Sigma}_a e'.$$

Two updated worlds are therefore indistinguishable when the agent cannot distinguish either their original worlds or their events.

The updated plausibility preorder is

$$(w, e) \leq'_a (w', e') \quad \Longleftrightarrow \quad \big(e <^{\Sigma}_a e' \wedge w \sim_a w'\big) \ \vee \ \big(e \simeq^{\Sigma}_a e' \wedge w \leq_a w'\big).$$

As the update is anti-lexicographic the ordering over events take precendence over the ordering over worlds. When two events are equiplausible, the previous ordering is left unchanged.

The resulting relations are passed back into the `EPM` constructor, which validates the updated model. Consequently, intermediate models in multi-step scenarios must satisfy the same well-formedness conditions as the initial model.

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4.5 Model Checking and Gold Answers

The recursive function `evaluate` determines whether a formula holds at a given world. Atomic propositions are checked against the world's valuation, and Boolean formulas are evaluated component by component. The evaluation rules for knowledge, belief, and conditional belief are given below.

$$M, w \models K_a\varphi \quad \Longleftrightarrow \quad M, v \models \varphi \text{ for every } v \in \{u \in W \mid w \sim_a u\},$$
$$M, w \models B_a\varphi \quad \Longleftrightarrow \quad M, v \models \varphi \text{ for every } v \in \mathrm{Min}_{\leq_a}\big(\{u \in W \mid w \sim_a u\}\big),$$

and

$$M, w \models B^{\psi}_a\varphi \quad \Longleftrightarrow \quad M, v \models \varphi \text{ for every } v \in \mathrm{Min}_{\leq_a}\big(\{u \in W \mid w \sim_a u\} \cap \{u \in W \mid M, u \models \psi\}\big).$$

Knowledge checks every world that the agent considers possible. Ordinary belief checks only the most plausible worlds in the agent's information cell. Conditional belief first restricts that cell to worlds satisfying ψ and then checks the most plausible remaining worlds. Ordinary belief is the special case

$$B_a\varphi \equiv B^{\top}_a\varphi.$$

Because evaluation is recursive, the same rules apply to nested formulas. For example, evaluating $B_aB_b\varphi$ first selects the worlds that agent a considers most plausible. The inner formula $B_b\varphi$ is then evaluated from each of those worlds.

If

$$\{u \in W \mid w \sim_a u\} \cap \{u \in W \mid M, u \models \psi\} = \varnothing,$$

then there are no minimal ψ-worlds to check. The conditional-belief formula is therefore vacuously true.

The function `holds` evaluates a formula at the actual world w^*. For each benchmark question, answer choices are associated with formulas. The function `gold_choice` evaluates these formulas using `holds` and selects the unique true answer. If no choice is true, it selects "no belief" when this is a valid choice. Syntactic-control questions use an explicitly human-assigned answer label, because their answers are determined by the narrative rather than the formal DEL model.

4.6 Validation

Both `EPM` and `ActionModel` validate their inputs when they are constructed. For each agent, the kernel checks that the indistinguishability cells form a genuine partition: no cell is empty, no element occurs in two cells, and every world or event is covered.

The kernel also checks that each plausibility relation is: reflexive; transitive; coherent with the indistinguishability partition, so that comparable elements belong to the same cell; and connected within each cell, so that any two elements in the same information cell are comparable.

The models are finite, so well-foundedness follows automatically and does not require a separate check.

Separate tests reproduce the central worked examples from Baltag and Smets [2008], including fair-game observation, fully private observation, and successful lying.

5 From DEL Models to Benchmark Items

5.1 Formal Scenario Instantiation

Each benchmark is initiated as a formal DEL construction. First, the prior coin model is instantiated by fixing the set of participating agents, the set of possible coin states, the actual coin state, and the initial plausibility ordering. This gives the initial epistemic plausibility model from which all the following questions are evaluated. One or more canonical action models are selected from the kernel. Depending on the scenario, this can be a public observation, a private observation, a truthful announcement, a successful lie, a lexicographic upgrade, or an unreliable announcement. These actions are applied using the Action-Priority product update described in Section 4. The result is a final epistemic plausibility model that encodes the relevant information structure, which gives all relevant information on agents beliefs afters updating. This final model is the formal source of truth for the benchmark item. The purpose of this section is therefore procedural rather than foundational: the kernel supplies the semantics, while the scenario-construction pipeline combines those semantics into concrete evaluation cases.

5.2 Natural-Language Realization

Once the formal scenario is set, it is rewritten as a natural-language story that LLMs can understand. This narrative reflects the model but does not determine the gold answer. Its purpose is to give the LLM the same information structure as the formal DEL model. The text clearly states who observed each event, who knows which facts, who only believes something, who trusts whom, and whether an announcement is truthful, deceptive, soft, or unreliable.

This order is important because LLM evaluations can be affected by knowledge learned during training. If a task depends on real-world facts, the model might answer based on its training data instead of using the information in the prompt, which can cause conflicts between what the model remembers and the context given [Xu et al., 2024]. To avoid this, we use synthetic coin-game scenarios, do not include real-world entities or trivia, and make sure all important information is in the prompt. This is similar to other theory-of-mind benchmarks that use synthetic scenarios to reduce contamination and keep the intended knowledge state clear [Sileo and Lernould, 2023, Strachan et al., 2024].

5.3 Probe Construction and Gold Labels

Probe questions are defined after the final formal model exists. Each probe has a natural-language question, a fixed set of answer choices, a logical depth, a recorded nesting structure, and a DEL formula attached to each formula answer choice. For example, a depth-1 probe may ask what Bob believes about the coin, while a depth-3 probe may ask what Bob believes Alice believes about Bob's knowledge.

The gold label is generated by model checking. Each answer-choice formula is evaluated in the final scenario model at the actual world. If exactly one formula answer is true, that answer is gold. If none of the formula answers is true, the designated `No belief either way` choice is gold. If more than one answer would be true, or if no valid fallback exists, the probe must be rejected or revised. This constraint ensures that the benchmark never depends on an informal judgment about what the answer "should" be.

5.4 Diagnostic Role of Scenarios

The scenarios are not an unstructured collection of puzzles. Each scenario has a diagnostic role. Baseline scenarios check factual and simple modal behaviour, such as whether the model can track the actual coin state after a public observation. Diagnostic pairs contrast minimally different epistemic structures, for example public observation versus private observation, or truthful announcement versus successful lie. These pairs make it possible to identify cases where a model is matching surface facts rather than tracking the underlying information state.

Other scenarios increase difficulty in controlled ways. Cross-axis scenarios combine two sources of complexity, such as private observation followed by soft announcement. Degradation scenarios preserve a core phenomenon while increasing step count, agent count, or the amount of private information. Stand-alone modal stress tests target special constructions that do not fit neatly into the main axes. This organization makes failures more interpretable, as an error can be related to observation, deception, trust, asymmetry, logical depth, or surface-form sensitivity rather than being treated as a generic wrong answer.

5.5 LLM Evaluation Flow

The evaluation procedure treats each probe as an independent model call. The renderer combines the scenario narrative, the probe question, and the fixed answer choices into a prompt. The LLM then returns one answer, which is parsed into a selected label. That label is compared with the DEL-generated gold label for the probe. The resulting trial data can be aggregated at several levels. We report accuracy by model, prompting strategy, scenario, logical depth, action type, and matched pair. This retains both the aggregate benchmark view and the diagnostic structure needed to explain where a model's higher-order belief tracking breaks down.

6 Scenario Suite

The scenarios are designed as a controlled suite to test different aspects of LLM reasoning. All scenarios use the same coin-game framing, which keeps the surface setting stable while varying the epistemic structure. This lets us ask whether an LLM's error is tied to a particular logical operation, such as private observation or deceptive testimony, rather than to a change in topic or background knowledge. The simplest scenarios use action templates from Baltag and Smets directly, while the harder scenarios compose those templates into extensions designed to diagnose specific shortcomings [Baltag and Smets, 2008].

6.1 Design Principles

The main design principle is controlled contrast. Each scenario is intended to isolate one source of difficulty: who observed the coin, whether an announcement was trusted, whether the speaker was deceptive, or whether different agents have asymmetric information states. The suite therefore does not merely test whether a model can answer isolated questions; it tests if performance changes predictably when a single epistemic factor is changed.

The second principle is compositionality. Scenarios are built from the action models in the kernel, including fair-game observation, private observation, successful lying, lexicographic upgrade, unreliable announcement, and public announcement. This makes the suite extensible: a new scenario can be added by composing existing action models, writing a matching narrative, and attaching probes to formulas in the final model.

6.2 Difficulty Axes

The core suite is organized along four difficulty axes. The observation axis begins with `coin_public_peek_v1`, where Bob observes the coin while Alice watches, and moves to `coin_private_peek_v1`, where Bob sees the coin without Alice knowing about it. The hard variant, `coin_double_peek_v1`, has both Bob and Alice privately peek in turn. This axis tests whether a model can distinguish direct knowledge from another agent's mistaken belief about who has knowledge.

The deception axis contrasts honest and dishonest communication. `coin_truthful_announcement_v1` uses a truthful announcement after Bob privately observes the coin, while `coin_successful_lie_v1` keeps the private observation but makes Bob lie to Alice. The harder `coin_lie_then_peek_v1` then gives Alice a later fair-game observation, testing whether the model can update a deception structure after new public evidence becomes available.

The trust axis varies the strength of testimony. In `coin_unreliable_announcement_v1`, an untrustworthy source claims tails and the agents' beliefs should not shift. In `coin_soft_announcement_v1`, a bystander gives a defeasible statement that shifts belief without producing knowledge. In `coin_conflicting_claims_v1`, Bob and Charles make incompatible claims, turning testimonial trust into a conflict between speakers.

The asymmetry axis increases the number of agents and the unevenness of their information states. `coin_successful_lie_3agent_v1` extends the successful-lie structure to Alice and Charles as trusting listeners. `coin_witness_deception_v1` then changes the structure: Bob and Charles observe the coin before Bob lies, so Charles witnesses the deception while Alice remains deceived. This makes the scenario hard not because the base fact is obscure, but because the agents' perspectives diverge.

Axis	Simple	Medium	Hard
Observation	`public_peek`	`private_peek`	`double_peek`
Deception	`truthful_announcement`	`successful_lie`	`lie_then_peek`
Trust	`unreliable_announcement`	`soft_announcement`	`conflicting_claims`
Asymmetry	`successful_lie`	`lie_3agent`	`witness_deception`

Table 1: The four main difficulty axes in the scenario suite.

6.3 Diagnostic Pairs

Several scenarios are paired so that a small formal difference creates a different gold answer. The pair `coin_public_peek_v1/coin_private_peek_v1` tests whether the model notices that Alice observes Bob's peek in one case but not the other.
The pair `coin_truthful_announcement_v1/coin_successful_lie_v1` keeps the announcement format similar while changing whether Bob's statement matches his information. The pair `coin_soft_announcement_v1/coin_unreliable_announcement_v1` isolates the effect of defeasible trust. The pair `coin_successful_lie_3agent_v1/coin_witness_deception_v1` tests whether the model tracks Charles's special role as a witness rather than treating all non-speaking agents alike.

These pairs are important because they expose fact-matching failures. A model can answer the factual coin question correctly while still giving the same higher-order answer in two scenarios whose DEL models require different belief states. Matched-pair analysis therefore asks whether the model is sensitive to the epistemic structure, not merely to the most prominent fact in the story.

6.4 Cross-Axis and Degradation Scenarios

The suite also contains scenarios that deliberately combine or scale difficulties. `coin_peek_then_soft_v1` combines private observation with soft testimony: Bob knows the coin state, but Alice hears a defeasible suggestion that can affect what Bob believes about Alice's beliefs. `coin_trusted_lie_v1` tests trust and speaker knowledge by making Alice trust Bob's false claim while Bob's own belief remains tied to the underlying prior structure.

The degradation scenarios test whether models remain stable as the same basic phenomenon becomes structurally larger. `coin_private_peek_3agent_v1` adds Charles to the private-peek setting. `coin_successful_lie_3agent_v1` adds Charles to the successful-lie setting. `coin_double_peek_v1` introduces mutual private observation, and `coin_peek_then_announce_v1` increases step count by adding a public announcement of ignorance after a fair-game peek. Finally, `coin_public_peek_tails_v1` serves as a factual flip control, and `coin_moore_sentence_v1` is a standalone modal stress test involving a true announcement that becomes false by being announced.

Separately from the main suite, a conditional-belief scenario tests the B_a^{ψ} modality directly. A card is either marked or plain, and probes ask what Alice *would* believe were she to learn each fact, including the nested case of what Alice believes Bob would come to believe. It is reported separately in Section 8.8 as a proof-of-concept that the kernel's conditional-belief semantics (Section 4.5) is used end-to-end.

6.5 Representative Scenarios

`coin_public_peek_v1` is the simplest public-observation case. Bob sees the coin while Alice watches, so both agents can reason not only about the coin state but also about each other's knowledge of Bob's observation. `coin_private_peek_v1` changes only the visibility of the event: Bob sees the coin while Alice is absent. This creates the characteristic private-information structure where Bob knows the coin state, but Alice believes Bob has not gained that knowledge.

`coin_successful_lie_v1` is the central deception case. Bob privately observes heads and then tells Alice that the coin is tails. Alice trusts Bob, so she forms a false belief about the coin; Bob, however, knows the truth and also has beliefs about what Alice now believes about him. This scenario is the main test of whether the model can separate the content of an utterance from the speaker's own belief state.

`coin_soft_announcement_v1` tests defeasible testimony rather than hard knowledge. A bystander suggests tails, shifting beliefs without eliminating the heads-worlds from the model. This is useful because many LLM errors collapse all testimony into either knowledge or irrelevance, whereas the DEL model distinguishes trusted, soft, and unreliable updates.

`coin_conflicting_claims_v1` and `coin_witness_deception_v1` are representative hard cases. The first requires tracking incompatible testimony from Bob and Charles. The second requires tracking three perspectives at once: Bob's deception, Alice's false belief, and Charles's witness knowledge. These scenarios are intended to break surface-level strategies and force genuine higher-order belief tracking.

7 Benchmark Design

7.1 Probes and Gold Answers

Each probe follows the schema introduced in Section 5.3: an identifier, logical depth, nesting structure, natural-language question, fixed answer choices, and formula mappings for formula-bearing choices. The final benchmark contains 140 probes from 17 scenarios 123 of which are logical probes and the other 17 syntactic controls.

"Gold" answers for logical probes are computed by model checking, as described in Section 5.3. Scenario construction fails if more than one formula-bearing answer is true, or if no formula-bearing answer is true and no `No belief either way` fallback is available. Syntactic controls bypass formula evaluation as they carry a fixed gold label derived from the narrative text.

7.2 Diagnostic Pairs and Controls

We use the diagnostic pairs described in Section 6.3 and the degradation and baseline controls described in Section 6.4. The additional benchmark-level constraint is that the behaviour of the pair is checked programmatically. For shared regular probes, paired scenarios must have identical gold labels. We declare diagnostic probes for each pair, whose gold labels must be different. This approach is only used for logical probes.

Each scenario also includes one syntactic control. These probes are nested but do not test any logical understanding as they are designed to be pure text comprehension, to isolate errors based in the writing, rather than logical reasoning.

Finally, each scenario has three narrative variants that are purely syntactical paraphrases of the same core. The probes, answer choices, formulae, and gold labels remain fixed to allow us to test whether failures persist under surface rewording.

7.3 Metrics

We report overall logical accuracy with Wilson 95% confidence intervals, excluding syntactic controls unless stated otherwise. We also report accuracy by model, prompting strategy, scenario, action type, and logical depth. For diagnostic pairs, outcomes are classified as `both_correct`, `same_wrong`, or `other`; `same_wrong` indicates a fact-matching failure, where the model gives the same answer to both members of a pair whose gold labels differ. We also include metrics on syntactic control accuracy and robustness between seeds.

8 Results

We evaluated five LLMs, one per provider (`gpt-4o-mini`, `claude-haiku-4.5`, `gemini-2.5-flash`, `mistral-small-3.2-24b-instruct`, `qwen3-32b`), using two prompting strategies (direct and chain-of-thought). Output parsing succeeded on 4196 of 4200 calls (0.10% parse-failure rate), so parsing is not a material issue for the models.

8.1 Overall accuracy

Pooled logical accuracy among all models, strategies, and seeds is 81.1% (95% CI $[79.8, 82.3]$, $N = 3690$), with models being separated clearly in some instance (Figure 1). `gemini-2.5-flash` (87.7%) and `qwen3-32b` (86.7%) score the best, followed by `claude-haiku-4.5` (83.2%) and `mistral-small-3.2` (79.0%).`gpt-4o-mini` scores far below the others at 69.0%, which is outside the confidence intervals of the other four. Chain-of-thought prompting produces small and inconsistent shifts. It improves the score for `gemini-2.5-flash` (+3.0 pp) and `mistral-small-3.2` (+3.6 pp) but has no to even slightly negative impact for `qwen3-32b` (−0.6 pp) and `gpt-4o-mini` (−1.3 pp). In every case the direct and CoT intervals overlap, so we find no robust main effect of CoT on higher-order belief tracking at this scale. Table 2 summarises the per-model results used throughout this section.

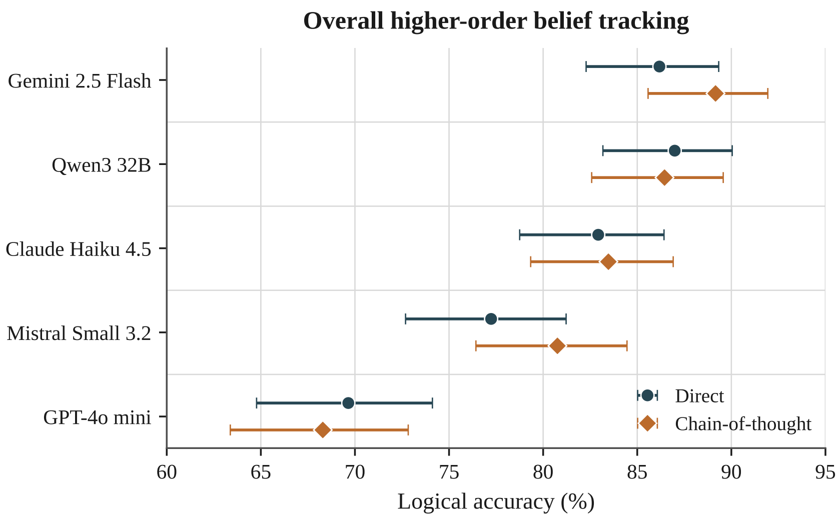

Figure 1: Overall logical accuracy by model and prompting strategy, with Wilson 95% confidence intervals. Models are ordered by pooled accuracy. Chain-of-thought and direct intervals overlap for every model.

Table 2: Logical accuracy (%) by model, pooled over seeds. *Direct* and *CoT* are per-strategy logical accuracy; *Overall* pools both strategies with a Wilson 95% interval. *Depth 3* is logical depth-3 accuracy. *Fact-match* is the share of matched-pair instances answered identically across both members of a diagnostic pair (a fact-matching failure; see Section 8.5). Models are ordered by overall accuracy.

Model	Direct	CoT	Overall (95% CI)	Depth 3	Fact-match
Gemini 2.5 Flash	86.2	89.2	87.7 [85.1, 89.8]	82.3	14.6
Qwen3 32B	87.0	86.4	86.7 [84.1, 89.0]	81.2	16.7
Claude Haiku 4.5	82.9	83.5	83.2 [80.3, 85.7]	79.2	22.9
Mistral Small 3.2	77.2	80.8	79.0 [75.9, 81.8]	63.5	13.5
GPT-4o mini	69.6	68.3	69.0 [65.5, 72.2]	41.7	28.1
Pooled	80.6	81.6	81.1 [79.8, 82.3]	69.6	19.2

8.2 Surface comprehension is not the bottleneck

The central result of DEL-Bench is that linguistic depth and logical depth dissociate. On the syntactic-control probes, which are linguistically nested to depth 3, accuracy is 99.4% ($N = 510$). On logical depth-3 probes, accuracy is significantly lower with 69.6% (95% CI $[65.3, 73.5]$, $N = 480$). Every model shows this gap (Figure 2), ranging from +17 to +18 pp for the three strongest models up to +36 pp for `mistral-small-3.2` and +58 pp for `gpt-4o-mini`, whose logical depth-3 accuracy collapses to 41.7% while its syntactic controls remain near ceiling. Because the controls share the surface form of the logical probes but require no belief revision, this gap isolates a failure of higher-order belief tracking rather than a failure to parse nested language.

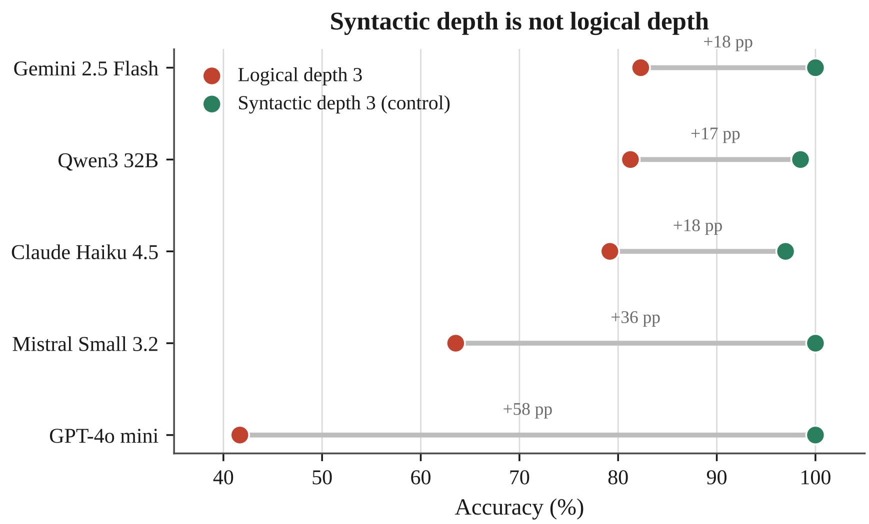

Figure 2: Syntactic depth is not logical depth. For each model, accuracy on logically deep depth-3 probes (red) versus syntactically deep but logically shallow control probes (green). All models handle the controls near ceiling; logical depth-3 accuracy is far lower.

8.3 Accuracy degrades with belief nesting

Pooled accuracy falls monotonically from depth 0 to depth 3: 99.4% (depth 0), 82.9% (depth 1), 77.2% (depth 2), and 69.6% (depth 3) (Figure 3). The apparent recovery at depth 4 (76.4%) is a composition artifact rather than a reversal of the trend: the depth-4 probes are drawn disproportionately from structurally simpler scenarios (for example, `coin_truthful_announcement_v1`, on which all models score 100%), whereas the hardest individual probes are concentrated at depth 3. The per-model curves show that the decline is steepest for the weakest model: `gpt-4o-mini` drops from 99% at depth 0 to 42% at depth 3, while `gemini-2.5-flash` retains 82% at depth 3.

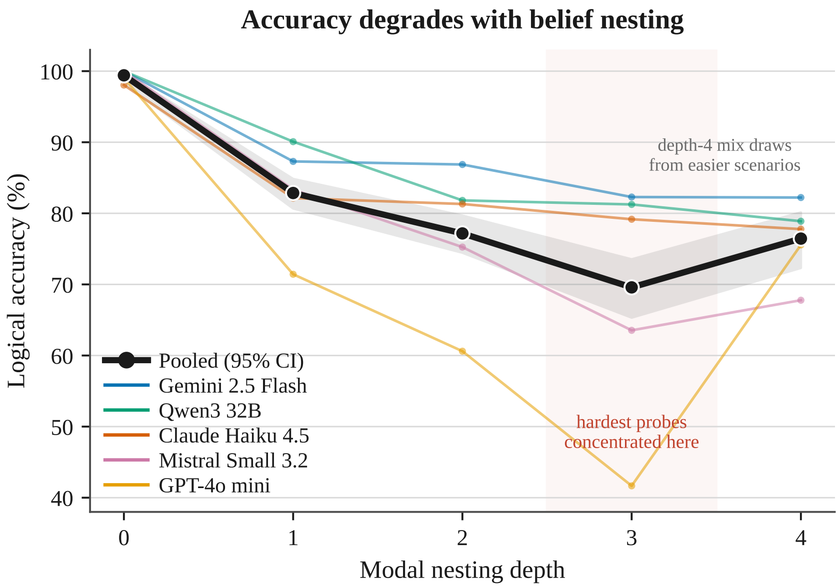

Figure 3: Logical accuracy versus modal nesting depth. The bold line is the pooled mean with a Wilson 95% band; thin lines are individual models. The depth-4 uptick reflects an easier scenario mix at that depth (see text).

8.4 Difficulty tracks epistemic structure

Accuracy per scenario differs widely from 55.2% to 100% (Figure 4). The easy end is dominated by baseline and single-update scenarios with low complexity. The the other side of the spectrum is dominated by the trust and deception composites: conflicting claims (55.2%), lie-then-peek (63.3%), and peek-then-soft (63.9%). Difficulty rises with the number of composed updates and with the divergence of agents' information states, not with surface topic, which is the same for all scenarios.

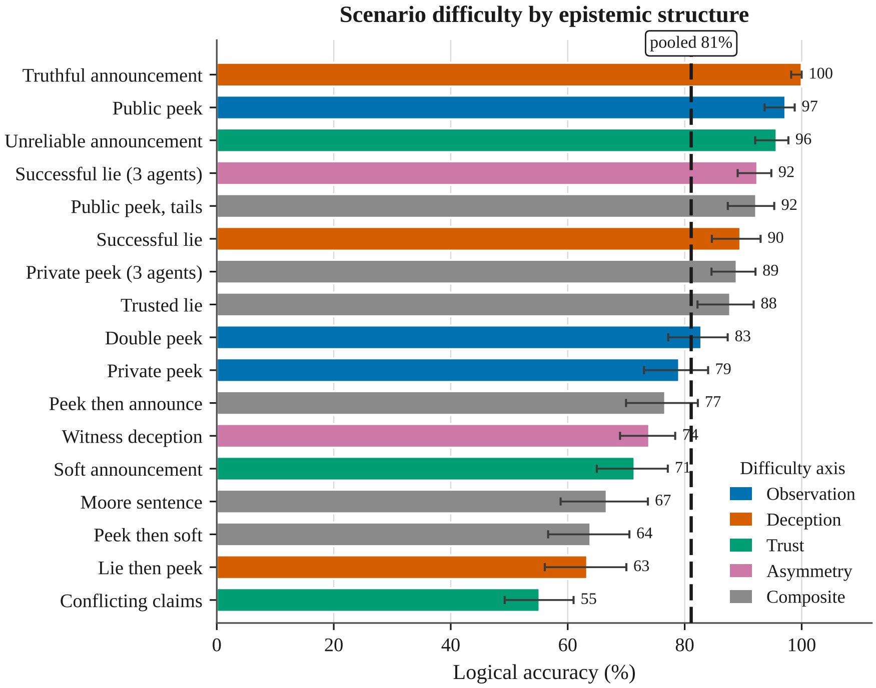

Figure 4: Per-scenario logical accuracy, pooled over models, strategies, and seeds, coloured by difficulty axis, with Wilson 95% intervals. The dashed line marks pooled accuracy (81%).

8.5 Matched-pair sensitivity

Aggregate accuracy can mask a specific failure mode in which a model answers from the most salient fact and ignores the epistemic structure that distinguishes two otherwise similar scenarios. We test this on the diagnostic pairs, restricting to shared probes whose gold answer differs between the two members, and classify each matched instance as *tracks structure* (correct on both), *fact-matching failure* (the same answer given to both members, which cannot be correct since their gold answers differ), or *other* (Figure 5). Fact-matching is hard for every model and is strongly negatively correlated with overall rank. It accounts for 14.6% of matched instances for `gemini-2.5-flash` but 28.1% for `gpt-4o-mini`. However, this is not uniform across the board. Haiku has a significantly larger percentage of fact checking than Mistral Small, but scores much higher in overall accuracy.

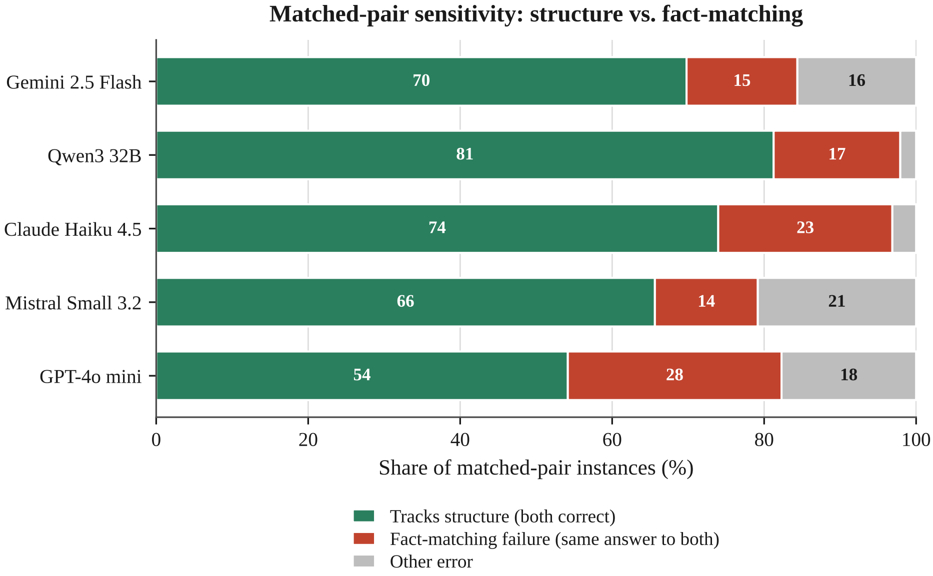

Figure 5: Matched-pair sensitivity on diagnostic pairs. Each bar decomposes matched instances into tracking the epistemic structure, fact-matching failure (identical answer to both members of a pair), and other errors.

8.6 Robustness to paraphrase

Because the gold labels are fixed by the DEL kernel and only the wording varies across seeds, we can isolate prompt sensitivity, which is a known issue for LLMs. We find promising results with per-model accuracy varying by at most 5.3 pp across seeds (Figure 6), with the strongest model the most stable (`gemini-2.5-flash`, 2.0 pp range) and weaker models somewhat more variable. Overall, this indicates that the measured differences reflect belief-tracking ability rather than sensitivity to a particular phrasing.

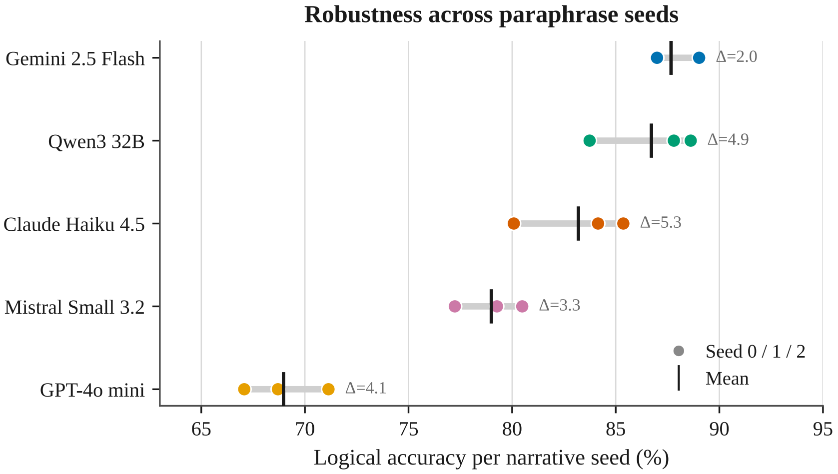

Figure 6: Logical accuracy across the three narrative paraphrase seeds. Dots are per-seed accuracies; the vertical mark is the per-model mean; Δ is the seed range.

8.7 Error analysis

To characterise how models fail, we decompose logical errors into three heuristic categories from the chosen and gold labels: a *belief-polarity flip* (an answer of the correct nesting form but opposite valence, e.g. "... believes heads" instead of "... believes tails"), a *no-belief confusion* (confusing a settled belief with the "no belief either way" fallback in either direction), and *other structural errors* (Figure 7). Polarity flips are the largest contributor for all models (43.9% to 59.2% of errors). This further illustrates the matched-pair finding. We see that models frequently recover the correct belief *structure* while assigning the wrong *content*. We treat this decomposition as an indication rather than exact, since it is computed from labels rather than from the formal models.

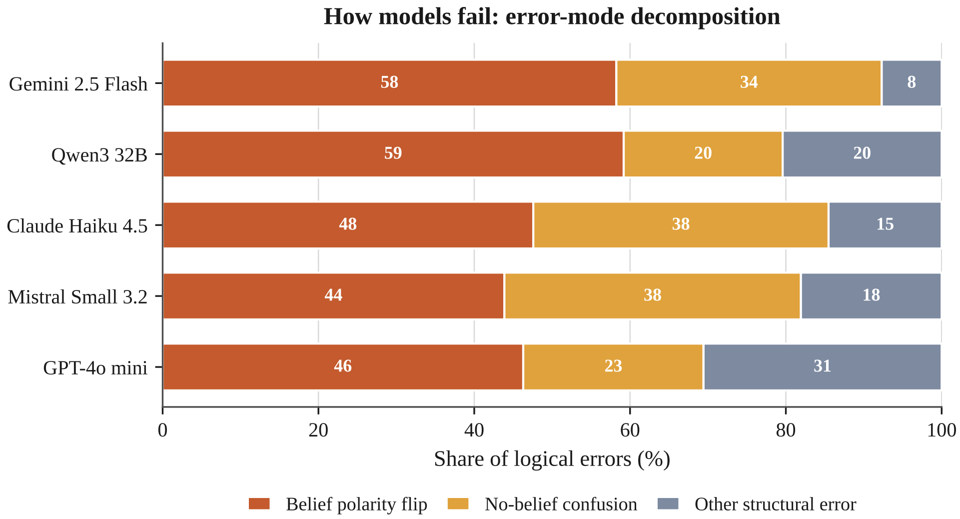

Figure 7: Decomposition of logical errors by mode. Belief-polarity flips dominate, suggesting that the dominant failure is wrong belief content on top of a recovered belief structure.

8.8 Conditional-belief extension

We additionally evaluate a conditional-belief scenario testing how models reason about hypothetical belief revision. This scenario is evaluated separately from the main scenario suite. Across five models, two prompting strategies, and three seeds, models correctly answer 88 of 90 conditional-belief probes (97.8%). First-order revision is handled with 100% accuracy. Performance falls slightly on the nested probe asking whether Alice believes that Bob would revise to tails after learning that the card is marked. This probe is answered correctly in 28 of 30 cases (93.3%) with both errors stemming from Mistral Small 3.2. These results are intended to show that the kernel supports conditional-belief evaluation, and amore comprehensive suite of conditional scenarios is needed we can draw any interesting conclusions.

Table 3: Conditional-belief accuracy on the standalone scenario, pooled over five models, two strategies, and three seeds.

Probe family	Correct	Accuracy
First-order conditional revision	60 / 60	100.0%
Nested conditional belief	28 / 30	93.3%
All conditional-belief probes	88 / 90	97.8%

9 Discussion

Three main findings are consistent across all five models. First, surface comprehension is mostly solved, but logical belief tracking is not: syntactic-control accuracy is at near maximum (99.4%).

Logical depth-3 accuracy paints another picture with mixed accuracy (69.6%). Second, accuracy is evidently correlated with nesting depth, with weaker models declining more. Third, many errors are due to fact-matching failures and belief-polarity flips. In this section we analyse the meaning of these results and classify the value provided by the benchmark.

9.1 Comprehension is not the bottleneck

The difference between syntactic and logical depth (Figure 2) is the clearest finding in this study. Since syntactic controls use the same nested language as logical probes but do not require belief revision, the gap cannot be blamed on trouble parsing complex sentences. The models can understand sentences like "a believes that b believes that c. . . ", but they struggle to determine the actual belief state when private observation, deception, or soft evidence changes the plausibility order. This closely mirrors earlier theory-of-mind studies which show the same decline with recusion depth [He et al., 2023] , and that understanding a mental state is something very different than applying it [Gu et al., 2024]. Our main contribution is to pinpoint this failure. Keeping the language the same and only changing the underlying knowledge structure, DEL-Bench shows that the problem is with belief tracking, not comprehension or test design [Strachan et al., 2024].

9.2 The diagnostic value of formal ground truth

The matched-pair analysis (Figure 5) is a clear indicator that the kernel generated ground truths are a crucial component. The experiments show that models answer the main factual question correctly but still give the same higher-order answer even though the scenarios actually require different belief states. That this is not a rare occurrence is highlighted by the density of these errors 13.5% to 28.1% of matched-pair cases. These errors do not appear in high-level accuracy metrics, since the scenarios only differ in who observed what. We avoid the informal judgments that can affect hand-written theory-of-mind tests by using formula checks at the actual world. [Sileo and Lernould, 2023]. We also match new standards for belief attribution in language models, by not only evaluating accuracy but also robustness [Herrmann and Levinstein, 2025]. By using synthetic and isolated scenarios that do not link to real world information, we avoid the contamination of pre-training [Xu et al., 2024].

9.3 Implications for multi-agent settings

The multi-agent risks discussed earlier in (Section 1), are confirmed by the errors identified. A fact-matching failure indicates that the model treats two different information states as the same, while a belief-polarity flip (Figure 7) is a mismatch of belief structure and actual beliefs of the agents. Both of these errors can mislead LLMs to be overly trusting of received information, which is a vulnerability that could be exploited by malicious agents [Hammond et al., 2025]. This links our observations to phenomena like sycophancy [Sharma et al., 2025], conformity to group opinions [Weng et al., 2025, Choi et al., 2025]. If an agent cannot tell the difference between what someone says and what they actually know, it cannot be relied on to defend against these risks.

9.4 Chain-of-thought has no robust effect

As observed in the results section, CoT has surprisingly little impact on accuracy(from +3.6 to -1.3 percentage points across models), with the confidence intervals for direct and CoT prompting always overlapping (Table 2). This could be an indicator of a representation error as models struggle to keep accuracy high when dealing with updates in text form. It also shows that the biggest challenge is not the reasoning process itself. Explaining intermediate steps does not help if the model cannot keep track of the plausibility order on which these steps depend.

This matches other work that supervises belief-update traces directly [Wu et al., 2025], and showcases model internal shortcomings that cannot be solved by prompting alone.

9.5 Limitations

There are several limitations to these findings. We only tested one model per provider, so we did not measure how results change with model size within each provider. The probe language only covers the K, B, and conditional-belief modalities; safe belief (Section 3.1) is not included. The error-mode breakdown (Figure 7 Finally, the depth-4 probes are not quite representative as they are mostly sampled for simpler scenarios compared to the depth-3 probes. This means that the increase in accuracy on depth-4 should not be taken at face value and the depth-3 probes are the most important metric to consider.

9.6 Future work

The current benchmark is limited in scope and serves as a proof of concept with early results. However, the Kernel works and can be scaled up and extended for future research. It could be interesting to consider additional scenarios next to the coin example chosen. Of course, more rigorous research would require the scope to be extended with more paraphrases, agents, and models. Since the kernel creates verified belief-update traces, it could also be modified to be used in training or verification of LLMs. It could serve as a parallel to trace-supervised theory-of-mind methods [Wu et al., 2025]. In the formal language, safe belief ($\Box$) is the only modality from the baseline paper that has yet to be included, and it is the next logical step. Overall, the modular design of the benchmark is very well suited for expansion.

10 Conclusion

This paper serves as a proof of concept. We have shown that dynamic epistemic logic, specifically the Action-Priority update of Baltag and Smets [2008], can serve as a reproducible ground truth for testing the ability of LLMs to track higher-order beliefs. To this end we developed a deterministic kernel that implements this framework and computes gold answers from an epistemic plausibility model when given formula-level translation of the natural-language scenario. We then evaluated several LLMs from different providers across a different scenarios spanning observation, deception, trust, and asymmetry. We find that this kind of stratified testing exposes weaknesses that aggregate accuracy may hide.

Our results give a good first indication of where these specific weaknesses may lie. When the same models are tested on sentences that are syntactically nested but logically shallow, they perform near ceiling at 99.4% accuracy, in contrast to the sharp drop in accuracy observed when genuine belief revision is required. This indicates that comprehension of the language is not the bottleneck. Additionally, as expected, accuracy degrades monotonically with depth of the nested beliefs, falling from 99.4% at depth 0 to 69.6% at depth 3. When performing the matched-pair analysis we found that in roughly 20% of cases (19.2% pooled, up to 28.1% for the weakest model), models give the identical answer to two scenarios whose gold answers differ only in epistemic structure, seemingly indicating the LLMs latch onto salient facts rather than actually tracking who believes what. This is reinforced by the most common error, belief-polarity flips, which account for 43.9% to 59.2% of errors across models, where models recover the correct belief structure (understanding the nested structure of the question) but assign it the wrong answer. Lastly and notably, chain-of-thought prompting offers no significant improvement on any of these tasks (shifting accuracy by only -1.3 to +3.6 percentage points depending on the model), which suggests the limitation lies in the representation of epistemic and doxastic states rather than it being a matter of reasoning effort that better prompting or higher token usage might fix.

In conclusion, current LLMs are prone to conflating what an agent says or what is most salient with what an agent actually knows. This is precisely the confusion that might make multi-agent systems vulnerable to deception and information asymmetry. We believe that this benchmark, once expanded and adapted to specific modes of multi-agent interaction, would enable detailed diagnosis of such weaknesses, and we hope to have demonstrated its promise here.

AI Disclaimer

For transparency we state our AI usage in the course of completing this research project.

- Gaining an overview of the current state of research in the field of DEL benchmarking.
- Coding assistance and annotation.
- Typesetting of the Latex Document.

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